

Question 2 $\Rightarrow$ Elevator Simulation

* Algorithm

① Start

② $\text{currentfloor} = 0$

③ $\text{count} = 1$

④ Input N

⑤ While $\text{count} \leq N$

 Input requested floor

 If

$\text{requested floor} > \text{current floor}$

 Print "Moving Up"

 Else If

$\text{requested floor} < \text{current floor}$

 Print "Moving Down"

 Else

 Print "Door Opening"

 End If

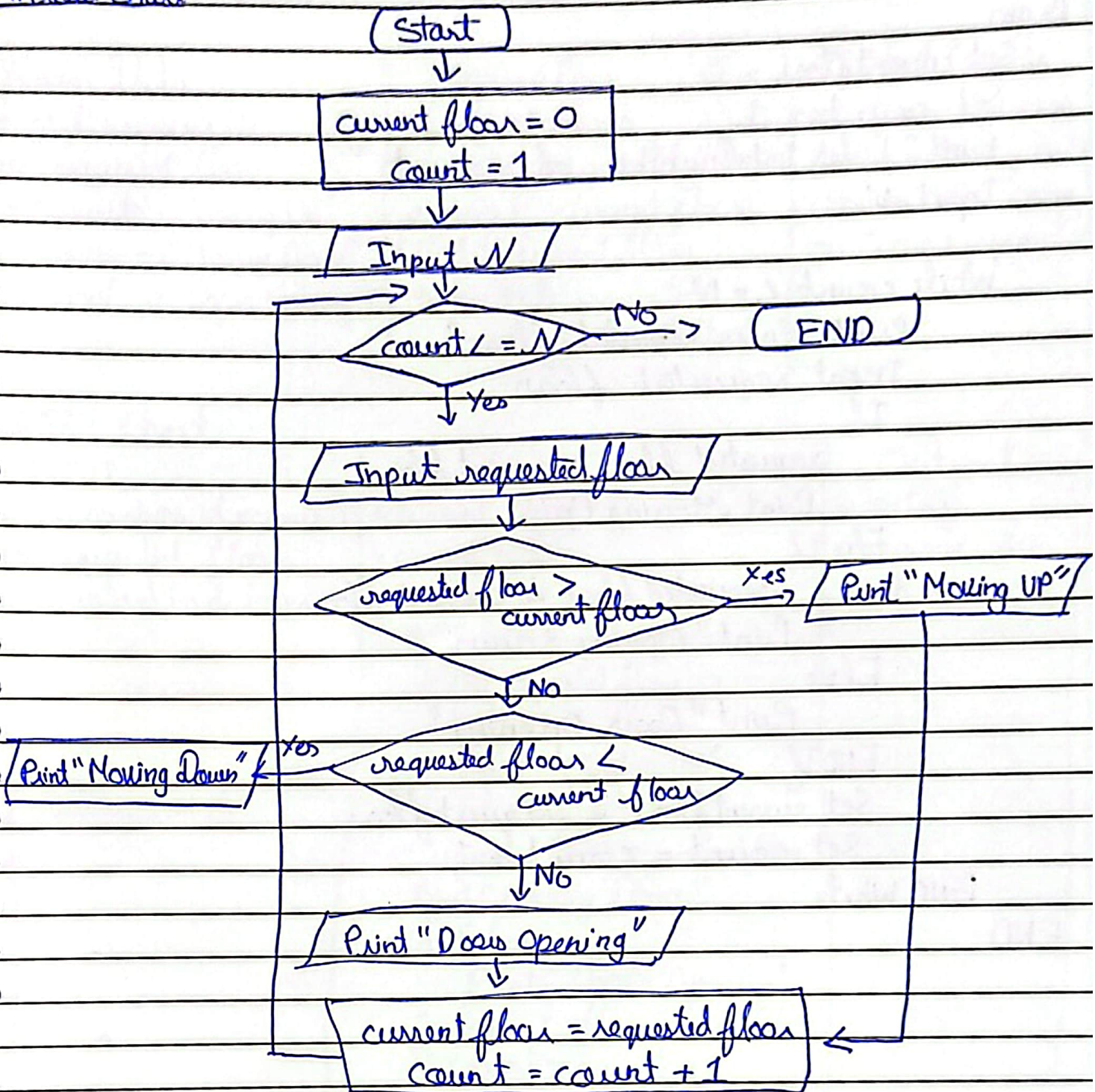
$\text{current floor} = \text{requested floor}$

$\text{count} = \text{count} + 1$

⑥ End While

⑦ End



*Flow chart

+ Pseudocode

Begin

Set current floor = 0

Set count = 1

Print "Enter total number of requests"

Input N

While count $\leq$ N

Print "Enter requested floor"

Input requested floor

If

requested floor $>$ current floor

Print "Moving Up"

Else If

requested floor $<$ current floor

Print "Moving Down"

Else

Print "Doors Opening"

End If

Set current floor = request floor

Set count = count + 1

End While

END



* Problem Analysis Chart (PAC)

Given Data	Required Process	Output Results
* Total requests (N)	* Loop N times	* Action message
* requested floor	* Compare requested floor against current floor	for each request
* Higher $\rightarrow$ "Moving Up"	* Update current floor to match requested floor after each stop	("Moving Up, Down or Doors Opening")
* Lower $\rightarrow$ "Moving Down"		
* Equal $\rightarrow$ "Doors Opening"		

* IPO Chart

INPUT	Processing	Output
* N (numbers of requests)	* current floor = 0	* Status
* requested floor (entered inside loop)	* count = 1	message for each move
	* Loop while count $\leq$ N	
	a) Input requested floor	
	b) If requested floor > current floor Print "Moving Up"	
	c) Else If requested floor < current floor Print "Moving Down"	
	d) Else Print "Doors Opening"	
	e) current floor = requested floor	
	f) count = count + 1	